Exquisitor at the Video Browser Showdown 2026: Temporal Queries Revisited

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Abstract. In today's data-rich world, multimedia content is produced at unprecedented rates, creating challenges for building systems that support evolving and often unknown information needs. Competitions such as the Video Browser Showdown (VBS) and the Lifelog Search Challenge (LSC) push researchers to develop systems that assist users in complex retrieval and analytical tasks. Exquisitor is an experimental, scalable multimedia retrieval system that integrates conversational search, relevance feedback, and metadata-based filtering to support exploratory and analytical search across large collections. Its development has been guided by the insights gained from participating in previous editions of VBS and LSC, evolving from a pure relevance feedback system to one that incorporates conversational interaction powered by large language models (LLMs). In this paper, we address two drawbacks observed in VBS 2025. First, tasks with temporal components highlighted the need for improved temporal querying. To address this, we propose a novel sequence-chain method combined with reciprocal rank fusion (RRF). Second, to enhance performance in question-answering tasks, we introduced in-video search to facilitate rapid content understanding.

Keywords: Multimedia retrieval · Temporal querying · Video browsing

1 Introduction

In today's ever-connected life, multimedia is created and gathered at a tremendous pace. One of the primary challenges of the multimedia research community is building experimental systems to support information needs (tasks) that we do not yet fully understand. Industry tends to focus on the tasks that can be (sometimes easily) solved with the technique of the day. These tasks recently changed from relatively simple textual search to more complex interactions supported by LLMs. Meanwhile, the space of potentially important retrieval and analytical

J. Lokoč et al. (Eds.): MMM 2026, LNCS 16415, pp. 245–251, 2026.
https://doi.org/10.1007/978-981-95-6963-2_27

tasks is much larger [7]. To address tasks that are not solved by industry, competitions such as the Video Browser Showdown (VBS) and the Lifelog Search Challenge (LSC) are useful, as they continuously evolve to focus on tasks that remain difficult to solve with existing approaches alone, requiring the retrieval systems to support the user in solving the tasks rather than solving them for the user. While the time pressure of the competitions can create an unnatural environment for users, participation in competitions is nevertheless valuable for multimedia researchers, forcing us to build and evaluate working systems.

Exquisitor is an experimental interactive and scalable multimedia information retrieval system, that combines text-based conversational search, relevance feedback, and metadata-based filtering to help users navigate large collections and locate items of interest for a given task. While Exquisitor is intended as a general retrieval and analytics system that can cover a large part of the exploration-search axis [13], we have used participation in VBS and LSC to drive the implementation, leveraging lessons learned from each competition to shape the trajectory of our explorative research.

In its initial participation in LSC 2019 and VBS 2020, the Exquisitor system was a pure relevance feedback system, only allowing the user to select positive and negative examples from an initial set of random images (or keyframes), and building and refining a model of the user's information needs as the interaction progressed. Metadata-based filters were soon added, as well as text-based search that allowed users to quickly find positive examples. With the advent of LLMs, the search was replaced by conversational search which, alone solves many of the competition tasks, while relevance feedback is now primarily used for the more difficult tasks that conversational search alone cannot solve. Along the way, the user interface has evolved to enable seamless transition between search and relevance feedback, making the system more friendly to novice users who are not familiar with the relevance feedback paradigm.

During VBS 2025, the latest incarnation of Exquisitor performed well, placing 4th out of 17 systems. In our analysis of the system's performance, however, we observed three primary deficiencies that we address in this paper.

– First, we observed that the tasks where we failed typically had a temporal component, such as: "A shot of a woman and a child at a forest clearing. The camera pans left to right, past a wooden cross in front, the woman and child walk to the left. There are some tree stumps, a big tree in the center. The woman is holding a white stuffed toy." (task vbs25-kis-t7). Clearly, the system lacks robust temporal search capabilities and more advanced video navigation tools are needed to handle complex multi-event queries. In VBS 2021, we experimented with handling queries with a temporal component by building separate models for each part and then fusing the results of the models. Building models of sufficient quality to find the relevant items in the intersection of the results was so time-consuming as to be impossible, however, leading to very limited success of this approach. To tackle this problem, we propose a new temporal querying method that finds sequence chains in videos matching the order of the selected queries, and uses the length of the

chain along with reciprocal rank fusion (RRF) to find relevant videos for the temporal task.
- Second, we observed that in question-answering problems, it can be easy to miss the answer, even when the correct video has been identified. We, therefore, add an in-video search that allows the user to gain quicker insight into such question-answering problems.
- Third, we used a high-dimensional index with Product Quantization (PQ) to reduce memory usage [12]. Although scalable, PQ degraded text-query and feedback accuracy. In the IViSE workshop [9] we used uncompressed embeddings since the dataset was smaller, which performed notably better.[1] To preserve accuracy with low memory use, we now adopt the disk-based eCP index [5], eliminating the need for compression.

The remainder of this paper is organised as follows. Section 2 goes over how temporal querying has been used in multimedia retrieval systems. Section 3 explains our proposed temporal query approach for Exquisitor, and Sect. 4 briefly describes the user interface and how it has been changed to better support video navigation. Finally, Sect. 5 concludes the paper.

2 Temporal Queries

In a multimedia analytics system, an array of multi-modal querying methods can be applied to find relevant items for a given task. These methods can be used individually or combined, typically through late fusion, to influence the final ranking of candidate items. Such approaches are highly effective for single-event retrieval. However, many multimedia collections contain temporally correlated items that are organized into (hierarchical) groups. Examples include video segments grouped under a parent video, or images and videos organized into albums corresponding to hierarchical temporal units such as days, months, or years, which may collectively span the entire dataset. The former type of grouping is prevalent in the datasets present in VBS [10], while the latter is characteristic of lifelogging datasets [3].

In such collection settings, information needs often involve multiple events occurring in sequence, whether driven by recalling known events or by exploratory curiosity to discover groups exhibiting certain temporal patterns. To support these complex temporally grounded information needs, many advanced retrieval systems have introduced temporal querying, enabling users to describe and search for multi-event sequences over time.

Temporal querying consists of two or more queries (these can be queries of single modality or multiple modalities), followed by a temporal constraint (e.g., before, after) and possibly a duration-based filter. A common strategy is to first obtain the result sets for each individual query, compute their intersection, as in videos containing results from all queries, and then apply the temporal constraints and filters [1,6,8]. It is typical for such approaches to be limited to two

[1] The IViSE workshop used the V3C1 subset of the V3C collection used in VBS.

queries, where matching results from the same video are paired. Constraints such as "q_2 happens after q_1" along with duration filters can be used to prune invalid pairs. A limitation of this independent-query approach is that each query must retrieve a sufficiently large number of results, especially in large scale collections. For instance, in Exquisitor this method was used with $k = 800$, yet it frequently exhausted the set of common videos and lacked an efficient mechanism to request additional relevant results without substantially increasing k.

Another effective strategy is to perform the subsequent queries within the temporal neighborhood of the segments or items retrieved by the initial query [2,4], followed by a ranking of the resulting sequences. This method typically assumes that the initial query represents the first event in the temporal sequence. Consequently, if new evidence later suggests that an earlier event should precede it, the search process may need to be restructured. Nevertheless, this approach offers greater control over the set of videos under consideration and can be easily extended by increasing the number of results retrieved for the initial query.

3 Proposed Approach

Our proposed temporal query approach is inspired by the idea of searching segments surrounding the initial query results. Consider a temporal sequence of queries $q_1, \ldots, Q_n$. We start by taking the initial query q_1 and retrieving the top r (by default, $r = 1000$) most relevant segments S_R. We then identify their videos and collect the segments that occur strictly after the matched q_1 segments(s); the union of all these "after" segments forms the pool of candidate segments S_C. Queries $q_2 ... q_N$ are then run against S_C in a single batched pass. From these results, chains are assembled per video by walking forward in time: a chain starts from its S_R segment and for each query q_i ($i \geq 2$), the highest-ranked segment from S_C is chosen that occurs after the previously chosen segment from q_{i-1}.

For ranking the query chains, we prioritize chain length first (longest match wins), and then apply Reciprocal Rank Fusion (RRF) as a rank-based tie-breaker over a chain's query-wise ranks. The fused score of a chain is determined by

$$RRF = \sum_i \frac{w_i}{k + r_i} \tag{1}$$

where $k > 0$ is a smoothing constant (by default, $k = 60$) that aims to prevent a single query severely outweighing the influence of the other queries, and w_i is an optional weight to specify query or hop importance (by default, $w_i = 1$). From the ranked list of chains, we select the highest-ranked chain per video and return the v distinct videos containing the top ranked chains (by default, $v = 50$).

Since videos can contain many overlapping chains, we use an overlap criterion of $IoU > \phi$ to identify highly redundant pairs, where IoU denotes the intersection-over-union of two chains and $\phi \in [0.0, 1.0]$ specifies the strictness of the overlap threshold. When two chains exceed this threshold, the longer chain is retained; if their lengths are equal, the chain with the higher RRF is kept.

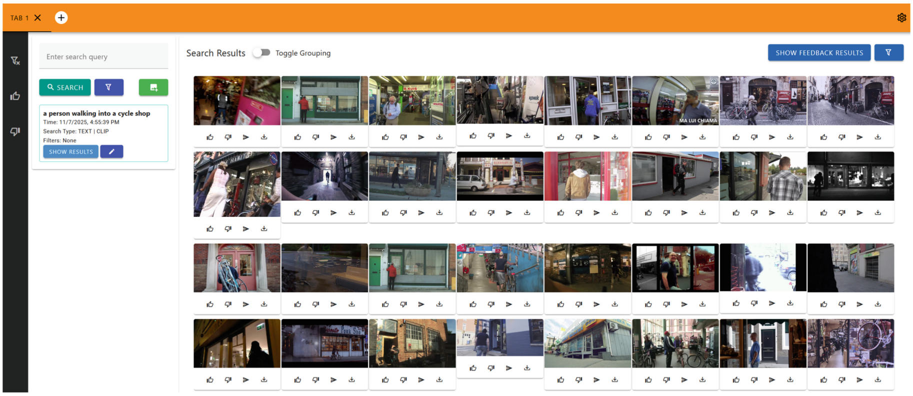

Fig. 1. Exquisitor's interface.

To support temporal querying through the Exquisitor interface, users can select queries from their query history list and run the temporal search. There will also be an option to include the current relevance feedback model, where the user has to select which query it comes after. The result of the queries will be displayed in Exquisitor's grouping view [11] (see Fig. 2), with the highest scoring chains at the top.

4 User Interface

Figure 1 depicts the user interface of Exquisitor which has four main components: the model bar on top of the screen, the left rail, the search pane, and the result view pane.

The **model bar** allows the user to open multiple tabs, where each tab has its own search history and relevance feedback model. Upon opening a tab the user is also prompted with which collection they want to use if multiple collections are available, which is the case for VBS with V3C, MVK, and LHE. The bar also has a settings button on the right side to rename the current tab, select which evaluation run to submit to and supply Q&A answers for the VBS competition. The **left rail** has buttons that open a panel to inspect the positive, negative, and excluded items. From these panels the user can adjust the relevance feedback model by removing previously labeled items or moving them to the other label, and reintroduce items from a video that had been excluded. The **search pane** has a text box at the top for search and a list below containing search history for the active model. The **result view pane** shows the results of a search, whether it is from the text search or from the relevance feedback model. The relevance feedback search is run by clicking the button labeled "Show Feedback Results" in the result view pane. Results are displayed either in a grid (Fig. 1) or in a grouped view if the "Toggle Grouping" switch is enabled (Fig. 2).

When a user clicks on any item (keyframe image representing a moment in the video) from the result view pane or the positive, negative, or excluded items

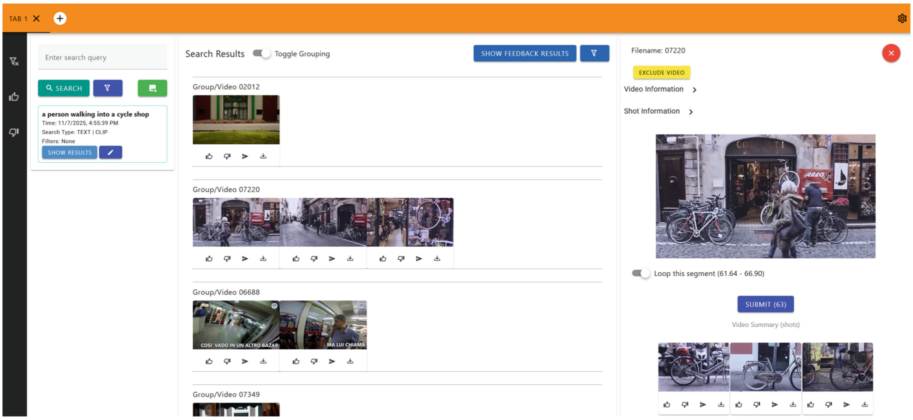

Fig. 2. Exquisitor's grouping view and the video summary panel. Each row shows the search result segments that belong to the same video in sequential order.

panels, they open the *video summary panel*. The summary panel plays the video from the selected moment in the default browser video player. Beneath the video player all the keyframes of the video are shown in a scrollable grid list to get a faster overview of the videos content, and they can be clicked on to jump to those moments as well.

Support for Question-Answering Tasks

In addition to supporting temporal search, the video summary panel of Exquisitor will be extended with an in-video search bar, enabling users to perform searches within a selected video. The goal of this feature is to improve fine-grained video navigation, particularly for complex or exploratory tasks such as question-answering or event localization. The in-video search dynamically re-ranks the video's segments according to their relevance to the new query, while preserving access to all keyframes.

5 Conclusion

Exquisitor is an interactive multimedia retrieval system, that offers a variety of ways for users to search their media collection. In this paper, we presented an extension to Exquisitor focusing on temporal retrieval, using sequential text queries and a relevance feedback model. Additionally, users can now search within a video to better support in-video exploration.

References

1. Amato, G., et al.: VISIONE 5.0: enhanced user interface and ai models for VBS2024. In: International Conference on Multimedia Modeling, pp. 332–339. Springer (2024)

2. Gia, B.T., et al.: NII-UIT at VBS2025: multimodal video retrieval with LLM integration and dynamic temporal search. In: International Conference on Multimedia Modeling, pp. 318–325. Springer (2025)
3. Gurrin, C., Smeaton, A.F., Doherty, A.R., et al.: LifeLogging: personal big data. Found. Trends® Inf. Retrieval **8**(1), 1–125 (2014)
4. Heller, S., et al.: Multi-modal interactive video retrieval with temporal queries. In: MultiMedia Modeling, pp. 493–498. Springer International Publishing, Cham (2022)
5. Khan, O.S., Guðmundsson, G.Þ., Jónsson, B.Þ.: The curious case of high-dimensional indexing as a file structure: a case study of eCP-FS. In: Similarity Search and Applications, pp. 303–311. Springer Nature Switzerland, Cham (2026)
6. Khan, O.S., et al.: Exquisitor at the video browser showdown 2021: relationships between semantic classifiers. In: Lokoč, J., et al. (eds.) MMM 2021. LNCS, vol. 12573, pp. 410–416. Springer, Cham (2021). https://doi.org/10.1007/978-3-030-67835-7_37
7. Lokoc, J., et al.: A task category space for user-centric comparative multimedia search evaluations. In: Proceedings of the International Conference on MultiMedia Modeling (MMM), pp. 193–204. Springer, Phu Quoc, Vietnam (2022)
8. Pantelidis, N., et al.: VERGE in VBS 2025. In: MultiMedia Modeling, pp. 355–362. Springer Nature Singapore, Singapore (2025)
9. Rossetto, L., et al.: Overview of the 1st international workshop on interactive video search and exploration. In: Proceedings of the Computer Vision and Pattern Recognition Conference, pp. 3673–3678 (2025)
10. Rossetto, L., Schoeffmann, K., Bernstein, A.: Insights on the V3C2 dataset. CoRR abs/2105.01475 (2021). https://arxiv.org/abs/2105.01475
11. Sharma, U., Khan, O.S., Rudinac, S., Jónsson, B.Þ.: Can relevance feedback, conversational search and foundation models work together for interactive video search and exploration? In: IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, CVPR Workshops, pp. 3740–3749. Computer Vision Foundation/IEEE, Nashville, TN, USA (2025)
12. Sharma, U., Khan, O.S., Rudinac, S., Jónsson, B.Þ.: Exquisitor at the video browser showdown 2025: unifying conversational search and user relevance feedback. In: International Conference on Multimedia Modeling, pp. 264–271. Springer (2025)
13. Zahálka, J., Worring, M.: Towards interactive, intelligent, and integrated multimedia analytics. In: 2014 IEEE Conference on Visual Analytics Science and Technology (VAST), pp. 3–12 (2014)